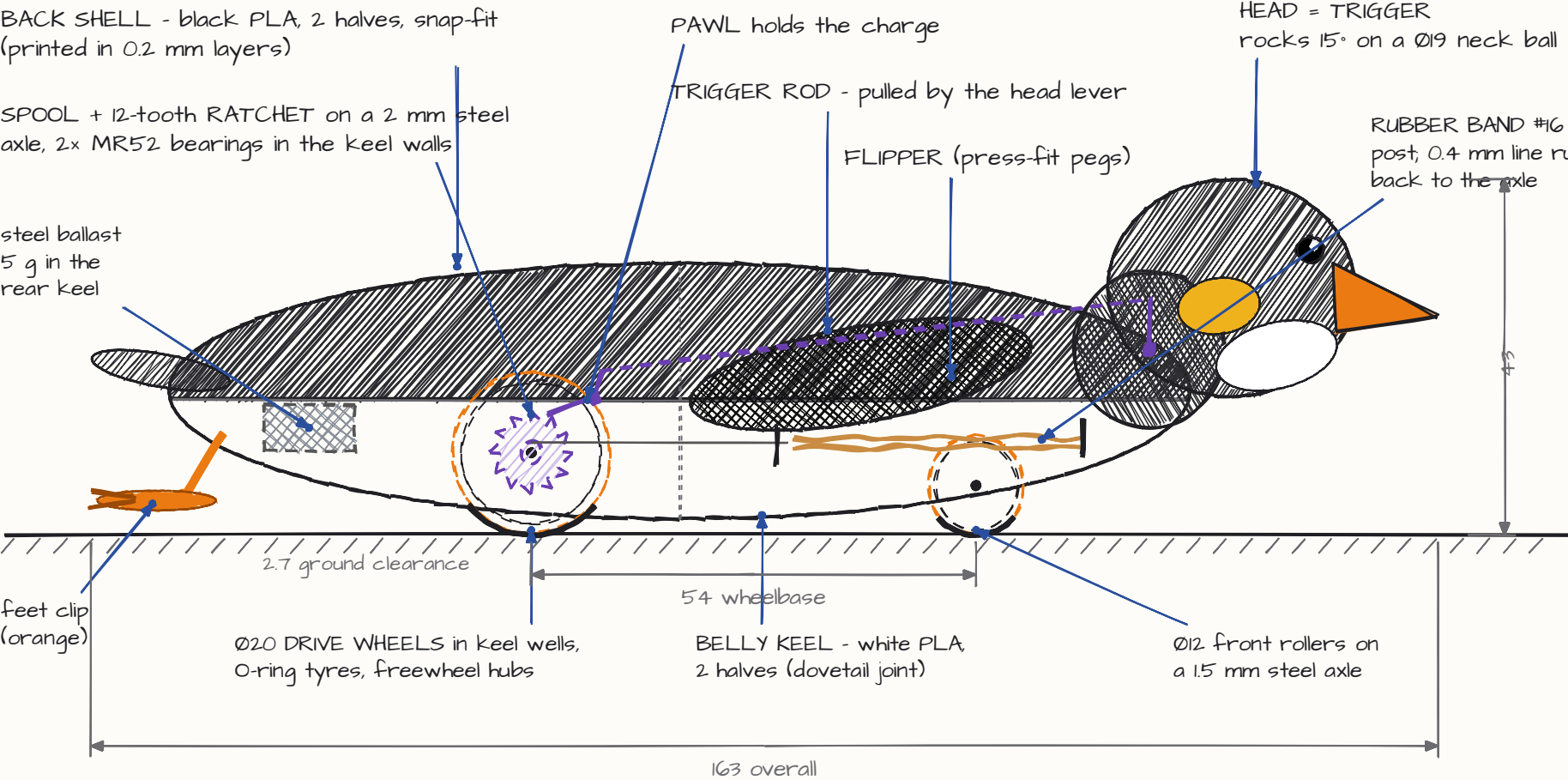
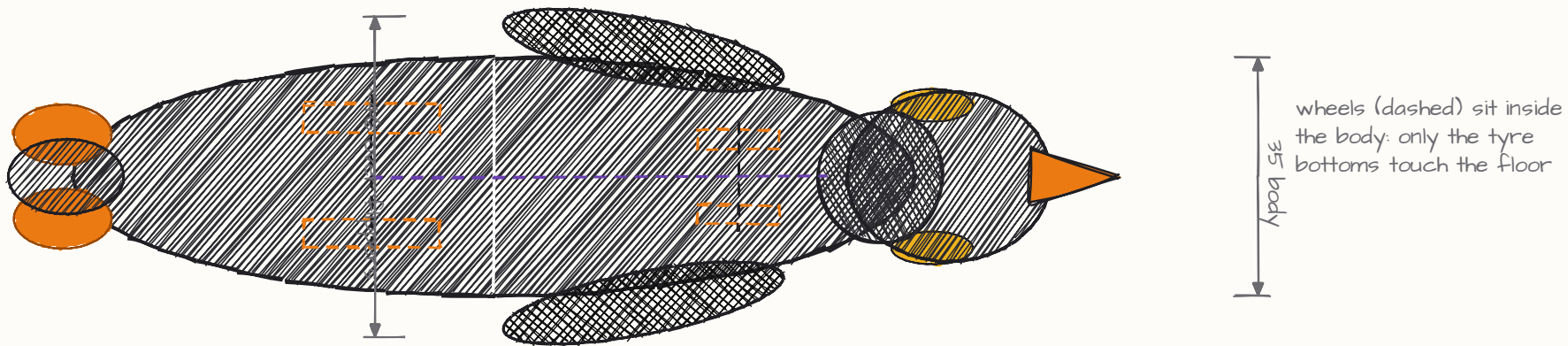


SIDE VIEW (dashed = hidden parts inside the body; purple = mechanism)



TOP VIEW

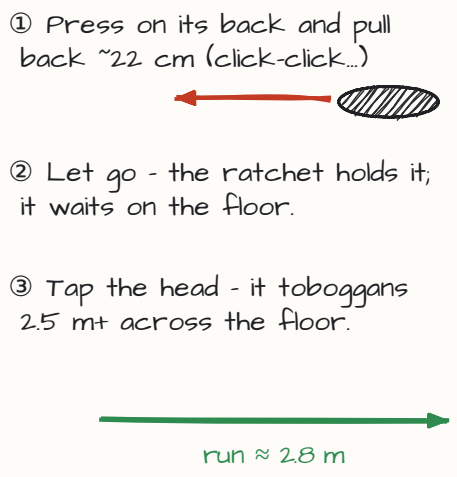


BIOMIMICRY



Emperor & Adélie penguins "toboggan" - belly on the ice, pushing with their feet. Copied here: low teardrop body (low drag), belly keel, rear "feet" drive, head-up pose.

HOW IT PLAYS



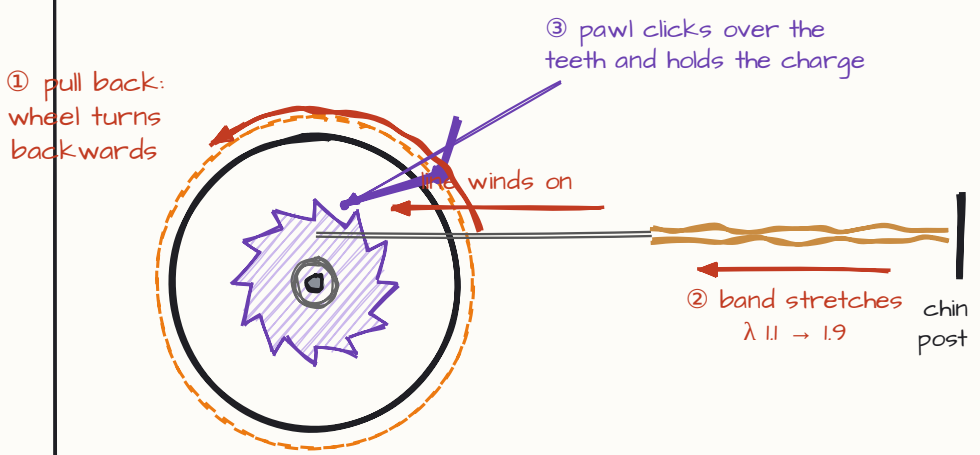
NUMBERS

- Mass 38 g, 79 % printed PLA
- CG 17 mm ahead of rear axle (68 % of weight on drive wheels)
- Band stores ≈ 31 mJ
- Predicted run 2.8 m, top speed 1.1 m/s
- No wheelspin above μ 0.64 (O-ring on oak ≈ 0.9)
- Bought: 2 + 1.5 mm steel rod, 4x MR52, 4 O-rings, #16 band, line, 5 g steel ballast

RULES CHECK

- ✓ ≥ 75 % printed PLA (79 %)
- ✓ Parts fit the egg; built toy is bigger than the egg
- ✓ Stores energy and waits until the head is tapped
- ✓ Trigger is not a removable pin - nothing comes off
- ✓ No energy stored when apart (band relaxed)
- ✓ Snap / press fits - no glue, no tools needed
- ✓ No bought toy parts or downloaded CAD

A ENERGY - PULL-BACK CHARGE



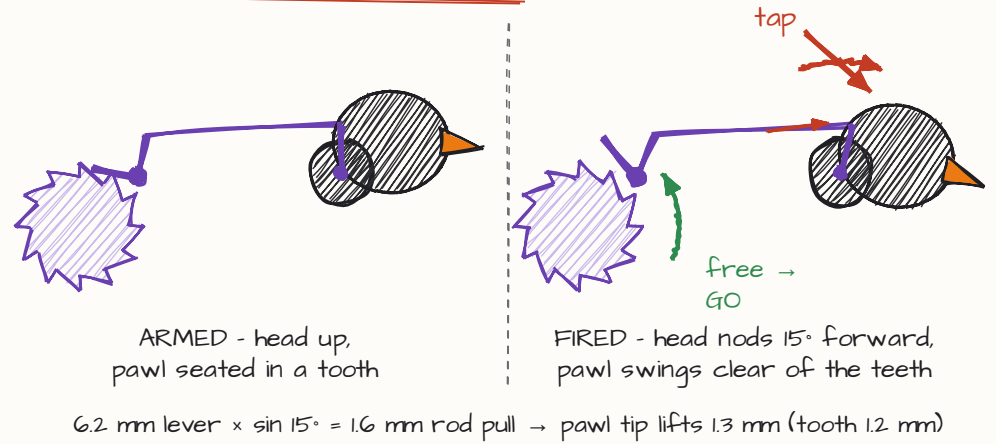
Stored: $E = GA_0L_0(\lambda^2/2 + 1/\lambda - 3/2) \approx 31 \text{ mJ}$ (G 0.45 MPa, A 2.56 mm², L_0 32 mm)

Needed for 2.5 m ≈ 26 mJ (rolling 16, front bores 8, bearings 2 mJ)

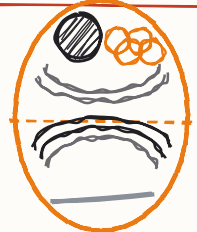
Over-pull? The band is fully wound at 22 cm, so the tyres just skid

Hubs = MR52 bearing + flexure pawl (freewheel): it coasts on after the line runs out instead of re-winding it.

B TRIGGER - TAP THE HEAD



C PACKS INTO THE EGG



All 23 parts nest inside the course Kinder Maxi egg (75 x 100 mm CAD); the shell halves stack like bowls.

Assembled 163 x 46 x 43 mm → bigger than the egg ✓

Concept 1: TUX TOBOGGAN

Theme: Biomimicry - tobogganing penguin

Toy type: Running toy (≥ 2.5 m)

Scale / units: mm · side view ≈ 1:31 on 11x17

Drawn by: _____ Signature: _____ Date: _____